Notes: Cohen, Coval, and Malloy (JPE, 2011)
Do Powerful Politicians Cause Corporate Downsizing?

Contents

1 Big picture

- **Question.** Does public-sector spending complement private-sector activity, or does it crowd out private investment and employment? The paper studies whether politically induced federal spending shocks change the behavior of firms headquartered in the recipient state.
- **Core empirical idea.** The paper uses changes in congressional committee chairmanships as a source of variation in state-level federal spending. Because chairmanships are largely seniority based, a state can receive a spending shock when one of its senators or representatives unexpectedly rises to a powerful committee position.
- **Main spending result.** When a state gains a powerful committee chair, its federal earmarks, transfers, and procurement contracts rise sharply. The paper estimates increases of roughly 40–50% in earmarks for some top Senate chair shocks, around 9–10% in total state-level federal transfers, and about 24% in government contracts.
- **Main firm result.** Firms headquartered in the affected state retrench. Capital expenditures fall, R&D falls, employment growth weakens, sales growth weakens in some specifications, and payouts to shareholders rise.
- **Economic mechanism.** The results are consistent with a local factor-market crowding-out channel. Government spending increases demand for state-specific labor and assets. Private firms then face higher local factor costs or weaker marginal products and scale back investment.
- **Why the channel is distinctive.** The empirical design reallocates federal spending across states rather than studying a national increase in federal spending. This makes standard national tax and interest-rate channels less central to the interpretation.
- **Key heterogeneity.** The negative firm response is strongest for firms with geographically concentrated operations, firms with high capacity utilization, and periods or states with little slack, such as low unemployment or high GDP growth.
- **Takeaway.** Powerful politicians bring more federal money to their states, but the average public firm headquartered in the recipient state responds by downsizing rather than expanding.

2 Incremental contribution of the paper

- **A new instrument for government spending.** Earlier fiscal multiplier studies often rely on aggregate time-series variation, defense buildups, or policy programs. This paper creates many state-level fiscal shocks using seniority-driven changes in powerful congressional committee chairmanships.
- **From aggregate output to firm decisions.** Instead of focusing only on GDP or employment aggregates, the paper studies firm-level capital expenditures, R&D, employment, sales, and payout decisions. This allows a more direct view of how private firms adjust.
- **Political economy meets corporate finance.** The paper connects political power, federal funds, and corporate investment behavior. The political allocation of spending becomes a tool for identifying real effects on firms.

- **A crowding-out mechanism beyond taxes and rates.** The paper shows that crowding out can arise through local factors of production even when the shock does not require higher national taxes or borrowing costs.
- **Evidence on local economic slack.** By showing larger effects when capacity utilization is high, unemployment is low, or GDP growth is high, the paper links fiscal crowding out to whether local resources are already heavily used.
- **Reversal evidence.** When a politician loses the chairmanship, some firm actions reverse. This strengthens the interpretation that the original effects are tied to the spending shock rather than permanent differences across states.
- **A caution for local fiscal policy evaluation.** The paper does not estimate a full national fiscal multiplier. It shows that for a state receiving politically allocated federal dollars, public spending can displace local private corporate activity.

3 Data and measurement

3.1 Congressional seniority shocks

- The key shock is a dummy equal to one when a senator or representative from a state becomes chair of an influential congressional committee.
- Chairmanships are defined using rankings of the most powerful congressional committees. For the Senate, examples include Finance, Veterans Affairs, Appropriations, Rules, Armed Services, Foreign Relations, Intelligence, Judiciary, Budget, and Commerce.
- For the House, examples include Ways and Means, Appropriations, Energy and Commerce, Rules, International Relations, Armed Services, Intelligence, Judiciary, Homeland Security, and Transportation and Infrastructure.
- The paper uses several versions of the shock: Top 1, Top 3, Top 5, and Top 10 committees; chair only versus chair plus ranking minority member; Senate versus House shocks; and drop variables for chair departures.
- In the baseline, a shock begins in the year of appointment and remains active for six years. The six-year window corresponds naturally to a Senate term and approximates the persistent nature of committee power.

Interpretation: The shock is meant to capture an increase in a state's ability to attract federal funds that is plausibly unrelated to short-run private-sector conditions in that state.

3.2 Federal spending data

- The paper collects congressional earmark data from Citizens Against Government Waste for 1991–2008.
- Many earmarks are not explicitly assigned to a state, so the authors hand-match more than 24,000 undesignated earmarks to states. For multistate earmarks, they split the amount equally across the designated states.

- The paper also uses broader federal transfer data from the Census Bureau's annual survey of state and local government finances for 1992–2007. It excludes category B79, which is nondiscretionary welfare spending such as Medicaid.
- A third spending measure is state-level government procurement contracts from Eagle Eye for 1992–2008.

Interpretation: Earmarks are the most discretionary and politically targeted measure, while transfers and contracts show that the seniority shocks affect broader flows of federal money.

3.3 Firm-level data

- The main firm sample contains 16,734 publicly traded firms from Compustat over 1967–2008.
- Firms are linked to politicians by the state of firm headquarters. The paper focuses on firms headquartered in the state receiving the political spending shock.
- Main outcome variables include capital expenditures scaled by lagged assets, R&D scaled by lagged assets, total payouts scaled by lagged assets, employment growth, and sales growth.
- Standard firm controls include Tobin's Q, cash flow scaled by lagged assets, and lagged leverage.
- Accounting variables are winsorized at the 1st and 99th percentiles.

Interpretation: The firm-level analysis is powerful because it uses audited public-company data and allows firm fixed effects, but it does not directly observe private firms or all establishments inside each state.

3.4 Geographic concentration and macro conditions

- The paper uses data on geographic concentration of firm operations to identify firms that operate only in one state.
- It uses capacity utilization data from the Federal Reserve to ask whether the effect is stronger when firms have little slack in capital use.
- It uses state unemployment, state real GDP growth, and U.S. real GDP growth to test whether crowding out is stronger in tighter economic conditions.

Interpretation: These variables are designed to distinguish a factor-market mechanism from a pure demand-stimulus story. If government spending mainly stimulates demand, the negative effect should not be concentrated in tight factor-market conditions.

4 Models and empirical specifications

4.1 Simple neoclassical mechanism

The paper motivates the empirical tests with a simple state-level neoclassical model. A representative agent has utility over consumption and leisure:

$$V(W_t) = \sum_t \beta^t U(C_t, L_t), \qquad 0 < \beta < 1. \tag{1}$$

The agent allocates wealth across consumption, taxes, federal transfers, in-state capital, and outside investment. In-state production uses capital and labor:

$$W_{t+1} = (W_t - C_t - G_t + E_t - K_t)(1 + r) + K_t^{\alpha}(1 - L_t)^{1-\alpha} + (1 - \delta)K_t. \tag{2}$$

The key first-order implication is that an unexpected increase in federal transfers E_t, if not fully offset by higher state spending or taxes, increases current wealth. The representative agent consumes more and takes more leisure. Less labor lowers the marginal product of in-state capital, so the optimal capital stock falls.

Prediction: A positive federal spending shock can reduce private employment, capital expenditures, output, and sales, while increasing payouts to shareholders if firms return unused funds.

4.2 State-level spending regression

The first-stage spending specification can be summarized as

$$\log(Spending_{s,t}) = \alpha_s + \tau_t + \beta Shock_{s,t} + \Gamma' X_{s,t} + \varepsilon_{s,t}, \tag{3}$$

where $Spending_{s,t}$ is earmarks, federal transfers, or government contracts in state s and year t. The specification includes state fixed effects, year fixed effects, and state-level controls such as population, lagged income growth, and unemployment.

Interpretation: The coefficient β asks whether a state receives more federal money when one of its politicians rises to a powerful committee position, relative to the same state's usual spending and to national year effects.

4.3 Firm-level outcome regression

The main firm-level reduced-form specification is

$$y_{i,t} = \alpha_i + \tau_t + \beta Shock_{s(i),t} + \Gamma' Z_{i,t-1} + \varepsilon_{i,t}, \tag{4}$$

where $y_{i,t}$ is a firm outcome such as capex/assets, R&D/assets, payouts/assets, employment growth, or sales growth. The firm is headquartered in state $s(i)$. The model includes firm fixed effects and year fixed effects.

Interpretation: The coefficient β compares the same firm's behavior before and after its state receives a political spending shock, controlling for national year shocks and standard firm characteristics.

4.4 Instrumental variables logic

The paper also discusses the structural problem that government spending is endogenous:

$$y_{i,t} = \alpha_i + \tau_t + \beta GovernmentSpending_{s(i),t} + \Gamma' Z_{i,t-1} + u_{i,t}. \tag{5}$$

Directly regressing firm investment on earmarks is not informative because spending may be sent to states with either weak or strong investment opportunities. Committee chairmanship shocks serve as instruments for the portion of government spending generated by political seniority rather than economic conditions.

Interpretation: The IV exercise shows that the politically induced part of earmarks predicts lower capex, while raw earmarks do not. This highlights the importance of breaking the endogeneity between spending and local private-sector conditions.

4.5 Heterogeneous effects

To test mechanisms, the paper estimates specifications such as

$$y_{i,t} = \alpha_i + \tau_t + \beta_1 Shock_{s(i),t} + \beta_2 Shock_{s(i),t} \times Tight_{i,s,t} + \Gamma' Z_{i,t-1} + \varepsilon_{i,t}. \tag{6}$$

The interaction variable can be high capacity utilization, low unemployment, high state GDP growth, or high U.S. GDP growth.

Interpretation: A negative β_2 means that firms retrench more when factors of production are scarce. This supports a crowding-out channel through labor and capital markets.

5 Identification and empirical design

5.1 Why chairmanships can identify spending shocks

- Congressional committee chairmanships are largely determined by seniority within the majority party on the committee.
- A new chair typically emerges because the old chair retires, dies, loses office, loses party control, or otherwise leaves the chair position.
- These events generally depend on political circumstances in other states or national party control, not on the economic conditions of the new chair's state.
- The resulting increase in federal spending is therefore plausibly exogenous to contemporaneous investment opportunities of firms headquartered in the new chair's state.

Interpretation: The design is a political natural experiment. It asks what happens when a state becomes politically more powerful for reasons that are not driven by local firm conditions.

5.2 What the design can identify

- The design identifies the effect of politically induced state-level federal spending shocks on public firms headquartered in the affected state.
- It is strongest for studying cross-state reallocations of federal spending rather than aggregate national fiscal expansions.
- It identifies corporate responses, not the total welfare effect for households, private firms, or the national economy.
- It can rule out national tax and interest-rate financing channels because the shock mostly shifts spending across states rather than raising total federal spending.

Interpretation: The paper is not estimating the full fiscal multiplier. It is identifying a local corporate response to a state receiving more federal dollars because of congressional seniority.

5.3 Threats and responses

- **Anticipation.** Firms might anticipate chair changes. The paper tests pre-shock coefficients and finds little evidence of pre-trends.
- **State-specific omitted shocks.** State and year fixed effects, firm fixed effects, controls, and state-level regressions reduce this concern.
- **House versus Senate.** The paper finds similar but smaller effects for House shocks, which is reasonable because House members may direct funds more narrowly to districts.
- **Large-state dominance.** The paper conducts state-level coefficient averaging so that a few large states do not drive the findings.
- **Headquarters versus operations.** Some firms headquartered in a state have operations elsewhere. The stronger results for single-state firms help address this concern.

Interpretation: The identification is not a randomized experiment, but the combination of seniority-based shocks, spending first stage, pre-shock tests, reversals, and mechanism tests makes the causal interpretation plausible.

6 Tables and figures

6.1 Table 1 and Figure 1: Summary statistics and the geography of earmarks

Read:

- Table 1 shows the main firm-level variables. Mean capex/assets is about 7.8%, mean R&D/assets is about 7.8%, and mean payout/assets is about 2.3%.
- The firm panel is large: capital expenditure regressions have about 169,000 firm-year observations.
- The shock variables show that seniority shocks are not rare, especially when the definition includes ranking minority members and more committees.

- The state-level panel shows average annual earmarks of about $139 million per state and average government contracts of about $2.3 billion.
- Figure 1 plots annual earmarks against state population. Population matters, but several smaller states, such as Hawaii, Alaska, Mississippi, West Virginia, and Alabama, are high-earmark outliers.

Economic message: Table 1 defines the empirical environment, and Figure 1 motivates the political-spending channel. Federal earmarks are not just a mechanical function of population; political power appears to matter.

Summary Statistics

	Mean	Median	Standard Deviation	Observations
	A. Firm-Level Annual Variables, Years 1967–2008, Firms = 16,734			
Capital Expenditures/Assets$_{-1}$	.078	.048	.108	168,975
Total Payout/Assets$_{-1}$	.023	.006	.044	154,832
R&D/Assets$_{-1}$	.078	.028	.134	86,870
ChgEmployees	.085	.026	.322	158,230
Cash Flow/Assets$_{-1}$	.036	.084	.242	151,482
Leverage$_{-1}$	.416	.399	.261	159,833
Tobin's Q_{-1}	1.822	1.230	1.826	153,348
Assets ($ millions)	2,845	194	26,667	168,970
Shock_Top1ChairOnly	.030	0	.171	168,975
Shock_Top1Chair&Rank	.032	0	.177	168,975
Shock_Top3ChairOnly	.044	0	.204	168,975
Shock_Top3Chair&Rank	.070	0	.255	168,975
Shock_Top5ChairOnly	.062	0	.242	168,975
Shock_Top5Chair&Rank	.118	0	.322	168,975
Shock_Top10ChairOnly	.098	0	.297	168,975
Shock_Top10Chair&Rank	.196	0	.397	168,975
Drop_Top1ChairOnly	.019	0	.136	168,975
Drop_Top3ChairOnly	.022	0	.146	168,975
Shock_Top1ChairOnly (House)	.037	0	.188	168,975
Shock_Top1Chair&Rank (House)	.100	0	.300	168,975
Shock_Top3ChairOnly (House)	.074	0	.261	168,975
Shock_Top3Chair&Rank (House)	.207	0	.405	168,975
Shock_Top5ChairOnly (House)	.113	0	.317	168,975
Shock_Top10ChairOnly (House)	.180	0	.384	168,975
	B. State-Level Annual Variables, Years = 1991–2008, States = 50			
Total Earmarks ($)	139,027,804	91,213,011	145,481,289	889
Ln(Total Earmarks)	18.17	18.33	1.25	889
State Population	5,327,111	3,665,228	5,811,533	889
Ln(State Population)	15.00	15.11	1.01	889
State Area (square miles)	72,694	56,276	87,559	889
Total State Govt. Transfers ($millions)	3,703.1	2,424.9	4,474.5	800
Log(Total State Govt. Transfers)	21.61	21.61	.88	800
Total Government Contracts ($ millions)	2,272.7	839.7	3,776.0	849
Log(Total Government Contracts)	20.10	20.55	2.20	849

Note.—This table reports summary statistics for the sample. Seniority shocks are defined as follows: Shock_Top1ChairOnly is a dummy variable equal to one if the senator (or representative) of a given state becomes chairman of the Senate Finance Committee (the House Ways and Means Committee); Shock_Top1Chair&Rank is equal to one if a senator becomes either chairman or the ranking minority member of the committee. The list of the top 10 most influential committees is from Edwards and Stewart (2006); for the Senate these committees are Finance, Veterans Affairs, Appropriations, Rules, Armed Services, Foreign Relations, Intelligence, Judiciary, Budget, and Commerce; for the House these committees are Ways and Means, Appropriations, Energy and Commerce, Rules, International Relations, Armed Services, Intelligence, Judiciary, Homeland Security, and Transportation and Infrastructure. Seniority shocks begin in the year of appointment and are applied for 6 years. All accounting variables are winsorized at the 1st and 99th percentiles of their distributions. The earmark data are from 1991–2008, the transfer data are from 1992–2007, and the contract data are from 1992–2008. All dollar figures are in 2008 dollars.

(a) Table 1: summary statistics

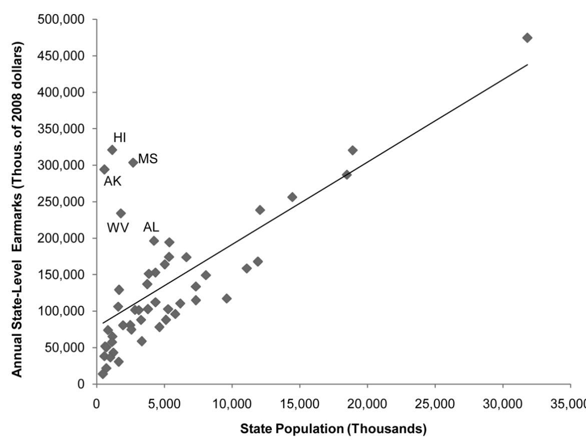

Fig. 1.—State-level annual earmarks versus population

(b) Figure 1: state earmarks versus population

6.2 Table 2: Average annual earmarks by state

Read:

- Table 2 ranks states by average annual earmarks during 1991–2008.
- California receives the largest total earmarks, but its rank is partly due to population size.
- Small states such as Alaska and Hawaii receive very high per-capita earmarks.
- The table also reports state-level averages of firm counts, capex, employment, and the Senate

and House shock variables.

- The shock variables show that political seniority exposure differs substantially across states.

Economic message: The table highlights the cross-state variation that the paper exploits. Some states receive unusually large earmarks relative to population, consistent with political representation and committee power shaping the allocation of federal funds.

TABLE 2
AVERAGE ANNUAL EARMARKS BY STATE

Earmark Rank	State	Annual Earmarks	Population	Population Rank	Per Capita Earmarks	Total Firms	Average Firms	Capex	No. Employees	Senate Shock	House Shock
1	CA	474,744,643	31,815,835	1	14.9	3,111	651.4	48,787.3	2,445.6	0	.3571
2	HI	320,872,527	1,159,883	41	276.6	25	8.5	596.2	21.1	.0976	0
3	TX	320,369,604	18,919,165	2	16.9	1,643	402.9	79,670.3	2,303.5	.1429	.3095
4	MS	303,468,441	2,708,937	31	112.0	58	12.3	318.6	20.0	.2381	.2857
5	AK	294,110,808	588,488	48	499.8	8	1.7	90.8	2.3	.4103	0
6	NY	286,856,506	18,483,456	3	15.5	1,872	472.1	60,058.6	3,525.9	.1429	.4286
7	FL	256,255,682	14,460,152	4	17.7	936	204.1	10,845.5	707.8	.1429	.2381
8	PA	238,567,184	12,081,349	5	19.7	675	192.7	18,970.4	1,192.7	.2857	.2857
9	WV	233,943,573	1,800,911	35	129.9	27	6.0	144.0	9.2	.2439	0
10	AL	196,160,333	4,243,844	23	46.2	92	24.9	1,494.8	93.9	.1429	0
11	WA	194,234,223	5,380,407	15	36.1	286	62.6	5,808.8	285.5	.1429	0
12	MO	174,110,006	5,356,142	16	32.5	232	70.3	7,881.7	670.2	0	0
13	VA	173,683,577	6,632,937	12	26.2	428	103.8	17,066.5	779.9	0	.1429
14	IL	167,702,049	11,924,948	6	14.1	728	208.5	39,578.9	2,837.9	0	.2857
15	MD	163,912,303	5,038,977	19	32.5	343	79.3	5,046.9	480.8	0	0
16	OH	158,257,270	11,100,128	7	14.3	521	169.3	20,429.4	1,804.9	0	.0238
17	LA	152,659,937	4,344,475	22	35.1	99	27.4	4,109.6	75.2	.1463	.3415
18	KY	151,019,194	3,863,533	24	39.1	92	23.9	2,022.4	213.5	0	.1429
19	NJ	149,090,487	8,072,269	9	18.5	916	233.1	36,555.6	1,618.6	.1429	0
20	SC	136,795,164	3,749,358	26	36.5	102	24.0	1,585.0	157.2	0	0
21	GA	133,430,550	7,332,335	11	18.2	465	108.2	15,364.1	742.6	.1429	0
22	NM	129,018,858	1,667,058	36	77.4	42	7.9	408.1	8.3	0	0
23	MI	117,090,209	9,616,871	8	12.2	315	100.3	37,643.1	2,104.6	0	.2857
24	NC	114,748,484	7,338,975	10	15.6	305	81.3	8,295.5	593.3	.0238	.1429
25	AZ	112,029,336	4,354,830	21	25.7	243	55.5	3,344.4	163.2	0	0
6	MA	110,318,321	6,182,761	13	17.8	889	210.4	8,153.8	680.6	0	.1429
7	NV	105,950,968	1,600,045	38	66.2	156	34.2	3,043.6	107.0	0	0
8	CO	102,720,845	3,797,828	25	27.0	645	120.3	10,592.9	246.2	0	0
9	TN	102,566,275	5,283,234	17	19.4	214	57.4	5,987.3	563.0	0	.2857
0	IA	101,493,645	2,851,540	30	35.6	90	25.8	1,182.4	66.4	.1429	0
1	OR	101,260,298	3,131,860	29	32.3	151	40.2	2,405.6	114.0	.2381	.1429
2	IN	95,933,350	5,812,322	14	16.5	204	56.3	3,792.9	215.5	0	.1429
3	WI	88,049,241	5,127,722	18	17.2	175	60.6	4,148.6	447.8	.1429	.1429
4	OK	87,815,500	3,289,147	28	26.7	214	41.1	7,156.0	60.6	0	0
5	AR	80,768,613	2,502,572	33	32.3	46	16.5	6,439.8	726.6	.1429	0
6	UT	80,426,189	1,963,000	34	41.0	150	30.8	1,541.1	112.3	.1429	0
7	MN	78,073,598	4,647,289	20	16.8	481	134.9	10,407.7	843.5	0	0
8	KS	74,640,287	2,582,996	32	28.9	120	29.5	3,798.3	137.8	.1429	0
9	MT	73,912,236	850,630	44	86.9	16	3.8	256.6	3.9	.1463	0
0	NH	65,175,744	1,176,241	40	55.4	78	19.8	520.0	41.4	0	0
1	CT	58,615,499	3,346,341	27	17.5	471	131.9	18,969.2	1,135.5	0	0
2	ID	57,388,189	1,150,351	42	49.9	36	10.5	1,516.7	99.0	.0952	0
3	ND	51,576,434	640,500	47	80.5	9	1.1	221.1	4.6	0	0
4	SD	50,399,507	725,424	46	69.5	17	5.3	279.8	16.5	0	0
5	ME	42,966,093	1,250,043	39	34.4	29	8.9	646.8	20.5	0	0
6	VT	38,018,251	585,793	49	64.9	18	5.6	109.5	4.0	.1463	0
7	RI	36,205,932	1,024,572	43	35.3	50	14.5	1,222.8	174.0	0	0
8	NE	30,339,541	1,644,824	37	18.4	56	13.6	2,944.2	164.7	.1463	0
9	DE	21,534,917	728,338	45	29.6	55	14.9	5,450.7	191.6	.1429	0
0	WY	13,695,506	472,503	50	29.0	17	3.1	13.5	.5	.2857	0

NOTE.—This table reports average annual earmarks by state for the period 1991–2008. Earmark figures are in 2008 dollars. Population figures for ach state are obtained from the 1990 and 2000 censuses. Total firms, average number of firms per year, average total capital expenditures per year in millions of 2008 dollars), and average total corporate employees per year (in thousands) are from Compustat and are yearly averages by state over ne full sample period (1967–2008). The shock variables are for the Shock_Top3Chair&Rank specification and are averages by state over the full ample period (1967–2008).

(a) Table 2, part 1: states ranked 1–25

(b) Table 2, part 2: states ranked 26–50 and notes

6.3 Table 3: Seniority shocks and federal spending

Read:

- Table 3 is the first-stage evidence that seniority shocks increase federal spending.
- In the earmark regressions, top committee chair shocks have large positive coefficients. The top Senate chair estimates imply roughly 40–50% higher earmarks.
- The effect weakens as the shock definition includes less powerful committees, but it remains positive and statistically significant across several definitions.
- The broader spending measures also move: total federal transfers rise by about 8.7%, and government procurement contracts rise by about 23.5% for the top chair specification.
- State and year fixed effects are included, so the estimates compare a state to itself over time while absorbing national spending cycles.

Economic message: This table validates the instrument. Becoming chair of a powerful committee translates into economically large increases in federal money flowing to the politician's state.

TABLE 3
THE IMPACT OF SENIORITY SHOCKS ON STATE-LEVEL EARMARKS, GOVERNMENT TRANSFERS, AND GOVERNMENT CONTRACTS

	EARMARKS									TRANSFERS	CONTRACTS
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Shock_Top1ChairOnly	.446*** (9.42)	.481*** (8.77)								.087** (2.54)	.235** (2.33)
Shock_Top3ChairOnly			.451*** (3.30)								
Shock_Top5ChairOnly				.426*** (3.67)							
Shock_Top10ChairOnly					.224** (2.49)						
Shock_Top1Chair&Rank						.330*** (3.30)					
Shock_Top3Chair&Rank							.265*** (3.12)				
Shock_Top5Chair&Rank								.205*** (2.76)			
Shock_Top10Chair&Rank									.164** (2.35)		
Controls	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effects	Year	Year	Year	Year	Year	Year	Year	Year	Year	Year	Year
Fixed effects	State	State	State	State	State	State	State	State	State	State	State
Adjusted R^2	.76	.76	.76	.76	.76	.76	.76	.76	.76	.98	.88
Observations	889	839	839	839	839	839	839	839	839	791	790

(a) Table 3, part 1: earmark and spending coefficients

NOTE.—This table reports panel regressions of earmarks, transfers, and procurement contracts on Senate seniority shocks (defined as in table 1). The dependent variable in cols. 1–9 is ln(state-level annual earmarks); in col. 10 it is ln(total state-level federal government transfers) from the Census Bureau (excluding category B79, which consists of nondiscretionary spending on public welfare items, e.g., Medicaid); and in col. 11 it is ln(total state-level procurement contracts), drawn from the Eagle Eye database. The earmark data are from 1991–2008, the transfer data from 1992–2007, and the procurement data from 1992–2008. Control variables include ln(state-level population), the state-level average of ln(per capita income) over the past 6 years, and lagged values of state-level ln(per capita income growth) and state-level unemployment rates. Year fixed effects and state fixed effects are included in all regressions. All standard errors are adjusted for clustering at the state level, and t-statistics using these clustered standard errors are included in parentheses below the coefficient estimates.

* Significant at 10 percent.
** Significant at 5 percent.
*** Significant at 1 percent.

(b) Table 3, part 2: controls and notes

6.4 Table 4: Seniority shocks and corporate investment

Read:

- Table 4 studies capital expenditures scaled by lagged assets.
- The coefficient on the top Senate chair shock is negative and statistically significant. In the first specification it is about -0.012.
- Since average capex/assets is about 0.078, a coefficient of -0.012 implies a roughly 15% decline in capex for the representative firm.
- The effect remains negative across alternative shock definitions and with controls for Tobin's Q, cash flow, and leverage.
- The drop variables are positive: when a chairmanship is lost, firms partly restore capital expenditures.

Economic message: This is the main reduced-form corporate result. Politically induced federal spending is associated with lower private corporate investment, and the reversal after chair departure strengthens the spending-shock interpretation.

TABLE 4

The Impact of Seniority Shocks on Corporate Investment, 1967–2008

	Dependent Variable: Capital Expenditures$_{i,t}/A_{i,t-1}$								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Shock_Top1ChairOnly	−.012***	−.009***							
	(3.46)	(3.14)							
Shock_Top1Chair&Rank			−.008***						
			(2.94)						
Shock_Top3ChairOnly				−.006***					
				(2.78)					
Shock_Top3Chair&Rank					−.006***				
					(3.57)				
Shock_Top5ChairOnly						−.005***			
						(2.64)			
Shock_Top10ChairOnly							−.003*		
							(1.95)		
Drop_Top1ChairOnly								.007**	
								(2.22)	

(a) Table 4, part 1: Senate shocks and capex

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Drop_Top3ChairOnly									.006**
									(2.11)
$Q_{i,t-1}$		.008***	.008***	.008***	.008***	.008***	.008***	.008***	.008***
		(12.33)	(12.33)	(12.32)	(12.34)	(12.29)	(12.33)	(12.33)	(12.32)
Cash Flow$_{i,t}/A_{i,t-1}$		.039***	.039***	.039***	.039***	.039***	.039***	.039***	.039***
		(9.40)	(9.40)	(9.40)	(9.40)	(9.41)	(9.40)	(9.40)	(9.40)
Leverage$_{i,t-1}$		−.116***	−.116***	−.116***	−.116***	−.117***	−.116***	−.117***	−.117***
		(31.54)	(31.53)	(31.51)	(31.50)	(31.46)	(31.54)	(31.42)	(31.41)
Adjusted R^2	.440	.501	.501	.501	.501	.501	.501	.501	.501
Observations	168,975	139,564	139,564	139,564	139,564	139,564	139,564	139,563	139,563

Note.—This table reports panel regressions of capital expenditures on Senate seniority shocks (defined as in table 1). All models contain firm fixed effects and year fixed effects. Controls for lagged Q, cash flow, and lagged leverage are included where indicated. All standard errors are adjusted for clustering at the state-year level, and t-statistics using these clustered standard errors are included in parentheses below the coefficient estimates.

* Significant at 10 percent.
** Significant at 5 percent.
*** Significant at 1 percent.

(b) Table 4, part 2: controls, reversals, and notes

6.5 Table 5: Alternative specifications for corporate investment

Read:

- Table 5 checks whether the capex result survives alternative specifications.
- House shocks also reduce capex, though the magnitude is smaller than for Senate shocks, consistent with House spending being more district-specific.
- The result holds among large firms, so it is not driven only by small companies.
- Excluding shocks in which the prior chair lost an election does not change the result.
- The pre-shock coefficient is essentially zero, indicating little evidence that firms reduce investment before the chairmanship change.
- In the earmark sample, directly regressing capex on earmarks is uninformative, but the IV-predicted value of earmarks from the seniority shock is negative and significant.

Economic message: The robustness evidence supports the identification strategy. The politically induced component of federal spending predicts lower investment, while raw spending is too endogenous to reveal the effect.

TABLE 5

ALTERNATIVE SPECIFICATIONS FOR THE IMPACT OF SENIORITY SHOCKS ON CORPORATE INVESTMENT

	Dependent Variable: Capital Expenditures$_{i,t}/A_{i,t-1}$							
	Full Sample (1967–2008)					Earmark Sample (1991–2008)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Shock_Top1ChairOnly (House shock)	−.004**							
	(2.26)							
Shock_Top3Chair&Rank (House shock)		−.003***						
		(2.74)						
Shock_Top3Chair&Rank (only large stocks)			−.005***					
			(3.14)					
Shock_Top3Chair&Rank (no lost elections)				−.006***				
				(3.44)				
Pre-shock					.001			
					(.37)			

(a) Table 5, part 1: House shocks, large firms, and pre-shocks

Shock_Top3Chair&Rank					−.009***	−.007***		
					(5.75)	(4.63)		
ln(Annual Earmarks)							.000	
							(.55)	
IV predicted value								−.008***
								(6.28)
Adjusted R^2	.501	.501	.611	.501	.393	.554	.510	.510
Observations	139,564	139,564	68,277	139,564	42,087	73,861	88,828	88,828

NOTE.—This table reports panel regressions of capital expenditures on House seniority shocks (from 1967–2008), earmarks directly (1991–2008), IV predicted values of earmarks (1991–2008), and various subsamples. Column 3 includes only stocks above the median lagged market capitalization in a given year in the regressions; col. 4 excludes all shocks in which the prior chairman lost his chair because he or she was defeated in an election/primary. Column 5 presents capital expenditure regressions similar to those in table 4 but that also include a variable called Pre-shock, which is a dummy variable equal to one in the 6 years prior to a shock. Column 6 runs the same regression as in col. 4 of table 4, but only for the 1991–2008 subperiod for which we have earmark data. Column 7 presents a regression of capital expenditures on ln(earmarks) directly. Column 8 presents the IV predicted value coming from an IV procedure using the first stage that regresses Shock_Top3Chair&Rank (Senate shock) on ln(earmarks), as in table 3. All models contain firm fixed effects and year fixed effects and controls for lagged Q, cash flow, and lagged leverage. All standard errors are adjusted for clustering at the state-year level, and t-statistics using these clustered standard errors are included in parentheses below the coefficient estimates.

* Significant at 10 percent.

** Significant at 5 percent.

*** Significant at 1 percent.

(b) Table 5, part 2: earmark sample, IV, and notes

6.6 Table 6: R&D, payouts, employment, and sales growth

Read:

- Table 6 expands the analysis beyond capex.
- Panel A shows that R&D/assets falls after seniority shocks. The paper interprets this as a reduction in investment in intellectual capital.
- Panel B shows that payouts/assets rise after seniority shocks. Firms return more money to shareholders when investment opportunities fall.
- Panel C shows that employment growth generally falls, especially for House shocks and in the earmark-period sample.
- Panel D shows weaker but still suggestive evidence that sales growth falls after spending shocks.
- As in Table 5, pre-shock coefficients are close to zero.

Economic message: The corporate response is broad. Firms do not merely delay one form of investment; they reduce capital and R&D spending, weaken hiring growth, and raise payouts.

TABLE 6

THE IMPACT OF SENIORITY SHOCKS ON CORPORATE R&D, PAYOUTS, EMPLOYMENT, AND SALES GROWTH, 1967–2008

	FULL SAMPLE (1967–2008)					EARMARK SAMPLE (1991–2008)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	A. R&D: Dependent Variable: R&D$_{i,t}/A_{i,t-1}$						
Shock_Top1ChairOnly	−.005***						
	(2.64)						
Shock_Top3Chair&Rank		−.003***	−.003***			−.003*	
		(3.04)	(2.82)			(1.66)	
Pre-shock			.000				
			(.12)				
Shock_Top1ChairOnly (House shock)				−.009***			
				(3.31)			
Shock_Top3Chair&Rank (House shock)					−.001		−.004**
					(1.28)		(1.98)
Adjusted R^2	.782	.782	.708	.783	.782	.777	.777
Observations	74,842	74,842	19,273	74,841	74,841	41,442	41,441
	B. Payouts: Dependent Variable: Payouts$_{i,t}/A_{i,t-1}$						
Shock_Top1ChairOnly	.003***						
	(4.15)						
Shock_Top3Chair&Rank		.001***	.001			.002**	
		(2.85)	(1.36)			(2.11)	

(a) Table 6, part 1: R&D and payout panels

Pre-shock			.000				
			(.34)				
Shock_Top1ChairOnly (House shock)				.001			
				(.84)			
Shock_Top3Chair&Rank (House shock)					.001***		.003***
					(3.59)		(4.84)
Adjusted R^2	.392	.392	.418	.392	.392	.412	.412
Observations	129,991	129,991	39,749	129,990	129,990	67,981	67,980
	C. ChgEmployees: Dependent Variable: (Employ$_{i,t}$ − Employ$_{i,t-1}$)/Employ$_{i,t-1}$						
Shock_Top1ChairOnly	−.009						
	(1.11)						
Shock_Top3Chair&Rank		−.011**	−.011***			−.020***	
		(2.41)	(2.67)			(2.64)	
Pre-shock			.000				
			(.01)				
Shock_Top1ChairOnly (House shock)				−.027*			
				(1.81)			
Shock_Top3Chair&Rank (House shock)					−.013***		−.025***
					(2.97)		(3.06)
Adjusted R^2	.135	.392	.086	.135	.135	.139	.139
Observations	168,267	129,991	41,577	168,265	168,265	89,702	89,702
	D. Sales Growth: Dependent Variable: (Sales$_{i,t}$ − Sales$_{i,t-1}$)/Sales$_{i,t-1}$						
Shock_Top1ChairOnly	−.015						
	(1.30)						
Shock_Top3Chair&Rank		−.014**	−.017***			−.030***	
		(2.31)	(3.11)			(2.90)	
Pre-shock			−.001				
			(.17)				
Shock_Top1ChairOnly (House shock)				−.054**			
				(2.09)			

(b) Table 6, part 2: employment and sales panels

TABLE 6 (*Continued*)

	Full Sample (1967–2008)					Earmark Sample (1991–2008)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Shock_Top3Chair&Rank (House shock)					−.024***		−.042***
					(3.49)		(3.55)
Adjusted R^2	.181	.181	.099	.182	.182	.187	.189
Observations	181,489	181,489	45,418	181,487	181,487	96,600	96,599

Note.—This table reports panel regressions of firm research and development (R&D), total payouts (cash dividends plus repurchases), firm-level changes in employment, and firm-level sales growth on Senate and House seniority shocks. Panel A reports results with firm-level R&D as the dependent variable, panel B reports results with firm-level payouts (cash dividends plus repurchases) as the dependent variable, panel C reports results with firm-level changes in employment as the dependent variable, and panel D reports results with firm-level changes in sales as the dependent variable. Each panel contains a specification that also includes a variable called Pre-shock, which is a dummy variable equal to one in the 6 years prior to a shock, plus results for the subperiod for which earmark data are available. All models contain firm fixed effects and year fixed effects. All standard errors are adjusted for clustering at the state-year level, and *t*-statistics using these clustered standard errors are included in parentheses below the coefficients.
* Significant at 10 percent.
** Significant at 5 percent.
*** Significant at 1 percent.

Table 6, part 3: continuation and notes

6.7 Table 7: Heterogeneous effects by economic conditions

Read:

- Column 1 restricts the sample to firms operating entirely in one state. The capex response is much larger than in the full sample.
- The interaction with high capacity utilization is negative, meaning firms with little capital slack reduce capex more strongly after a spending shock.
- The interaction with low unemployment is negative for employment, meaning crowding out is stronger when local labor markets are tight.
- Interactions with state real GDP growth and high state GDP are negative for capex and employment.
- Interactions with high U.S. GDP growth are also negative, though generally weaker than state-level interactions.

Economic message: The table is the key mechanism evidence. Government spending appears to crowd out private firms mainly when resources are scarce, consistent with factor-market competition rather than a pure demand expansion.

TABLE 7
Heterogeneous Effects of Seniority Shocks by Economic Conditions: Interactions with Utilization, Unemployment, State-Level Real GDP Growth, and U.S. Real GDP Growth

	Capex (1 State) (1)	Capex (2)	Employ (3)	Capex (4)	Employ (5)	Capex (6)	Employ (7)	Capex (8)	Employ (9)
Shock_Top1ChairOnly	−.031**	−.010***	.008	−.006***	.004	−.005**	.005	−.009***	.007
	(2.20)	(3.28)	(.85)	(2.91)	(.57)	(2.38)	(.60)	(4.31)	(.79)
Shock_Top1ChairOnly × HighUtil (sample period: 1980–2008)		−.014***							
		(2.95)							
Shock_Top1ChairOnly × LowUnemp (sample period: 1977–2008)			−.024**						
			(2.17)						
Shock_Top1ChairOnly × StateGDP (sample period: 1967–2008)				−.193***	−.411*				
				(3.70)	(1.93)				
Shock_Top1ChairOnly × HighStateGDP (sample period: 1967–2008)						−.014***	−.025***		
						(5.85)	(2.56)		
Shock_Top1ChairOnly × HighUSGDP (sample period: 1967–2008)								−.004*	−.023**
								(1.79)	(2.06)

(a) Table 7, part 1: geographic concentration and slack interactions

Iigh Utilization (sample period: 1980–2008)		.007***							
		(6.31)							
.ow Unemployment (sample period: 1977–2008)			.008**						
			(2.40)						
itate GDP (sample period: 1967–2008)				.250***	.411***				
				(27.28)	(11.33)				
Iigh State GDP (sample period: 1967–2008)						.006***	.012***		
						(13.84)	(7.83)		
Iigh U.S. GDP (sample period: 1967–2008)								.024***	.145***
								(14.04)	(21.88)
\djusted R^2	.730	.500	.227	.443	.136	.440	.136	.440	.135
)bservations	3,624	49,520	147,886	168,975	168,267	168,975	168,267	168,975	168,267

Note.—This table reports firm-level panel regressions of capex and employment on Senate seniority shocks and various interaction terms. As n tables 4 and 5, capex is firm-level capital expenditures divided by lagged assets, and employ is the change in number of employees. Column 1 ncludes only the sample of firms that have all operations in a single state. High Utilization is a dummy variable equal to one if the firm in question s above the median in terms of its capacity utilization rate. Low Unemployment is a dummy equal to one if the difference between the state-level ınemployment rate (in the state the firm is headquartered in) and the national unemployment rate is less than its historical average difference. itate GDP equals the real growth rate in the state's GDP (drawn from the BEA). High State GDP is a dummy equal to one if the difference between he state-level real GDP growth rate and the national real GDP growth rate is greater than its historical average difference. High U.S. GDP is a lummy equal to one if the national real GDP growth rate is greater than its historical average. All models contain firm fixed effects and year fixed ffects; standard errors are adjusted for clustering at the state-year level, and *t*-statistics using these clustered standard errors are included in parentheses below the coefficient estimates.
* Significant at 10 percent.
** Significant at 5 percent.
*** Significant at 1 percent.

(b) Table 7, part 2: GDP interactions and notes

7 Short summary

- The paper studies whether federal spending shocks stimulate or crowd out private corporate activity.
- The identification strategy uses changes in powerful congressional committee chairmanships, which are largely determined by seniority.
- These chairmanship shocks increase the recipient state's federal earmarks, transfers, and procurement contracts.
- Public firms headquartered in the recipient state reduce capital expenditures after the shock.
- They also reduce R&D, reduce employment growth in many specifications, and increase payouts to shareholders.
- The response reverses partly when the politician loses the chairmanship.
- There is little evidence of pre-shock behavior, which supports the timing assumption.
- The effect is strongest among geographically concentrated firms and in periods with little factor slack.
- The mechanism is not the classic tax or interest-rate channel. It is local crowding out through labor, fixed assets, and other state-specific factors.
- The paper's main contribution is to provide firm-level evidence that politically induced government spending can cause corporate downsizing.

8 Conclusion

8.1 Core conclusion

Cohen, Coval, and Malloy show that when a state gains a powerful congressional committee chair, the state receives more federal spending, but public firms headquartered in that state reduce private investment. The empirical pattern is more consistent with neoclassical crowding out than with a simple Keynesian stimulus story for public firms.

8.2 Economic intuition

The intuition is that federal funds can increase demand for local resources. If workers, construction inputs, government contractors, land, and specialized assets become more demanded by the public sector, private firms may find investment less attractive. If residents feel wealthier, they may also work less, lowering the marginal product of capital. Firms respond by reducing capex and R&D and by returning more cash to shareholders.

8.3 Incremental contribution revisited

The paper advances the literature by turning political seniority into an empirical instrument for local federal spending. It then connects that shock to corporate finance outcomes. This combination makes the paper more micro-founded than many aggregate fiscal studies and more policy-relevant than a purely political allocation study.

8.4 Implications for fiscal policy

The results do not imply that all government spending is bad or that fiscal policy cannot increase aggregate welfare. They show that, in a local setting, politically allocated federal funds can displace public-firm investment. This is especially important when factor markets are tight. Fiscal stimulus may be more likely to crowd out private activity when an economy has little slack.

8.5 Limitations

The study focuses on public firms and headquarters states. It may miss private firms, household consumption, and within-firm resource shifts across states. The paper also studies reallocations of federal funds across states rather than aggregate national spending shocks. For these reasons, the results should not be read as a complete estimate of the fiscal multiplier.

8.6 Takeaway

The enduring takeaway is that political power has real economic consequences. Powerful politicians bring federal dollars home, but the average public corporation in the recipient state responds by investing less, hiring less aggressively, and paying out more. The paper therefore adds a factor-market crowding-out channel to how we think about government spending and private-sector activity.